

1 Question (18 points)

Give state diagrams of NFA's with the specified number of states recognizing each of the following languages:

- $0^*1^*0^+$ (3 states)
- w contains the substring 0101 (5 states)

2 Question (18 points)

Let $B_{(m,n)} = \{x \mid x \text{ is a number in modulus } m \text{ that is a multiple of } n\}$. Show that for each $m, n \geq 1$, the language is $B_{(m,n)}$ is regular.

3 Question (18 points)

Write regular expressions for the following languages. Let $\Sigma = \{0, 1\}$.

- $\{w \mid \text{every odd position of } w \text{ is a } 1\}$
- $\{w \mid w \text{ contains at least two 0s and at most one } 1\}$
- $\{w \mid w \text{ contains an even number of 0s or contains exactly two 1s}\}$

4 Question (18 points)

Are the following languages regular? Prove or disprove.

- $A = \{1^n \mid n \geq 0\}$
- For a given $p \in \mathbb{N}$, $A = \{a^n b^m \mid n, m \geq 0, n + m = p\}$

5 Question (18 points)

Let $\Sigma = \{a, b\}$. For each $k \geq 1$, let C_k be the language consisting of all strings that contain an 'a' exactly k places from the right end.

- Describe an NFA with $k + 1$ states that recognizes C_k .
- Describe an DFA with 2^k states that recognizes C_k .